Error-Controlled Bayesian Localization for Passive Radar With Receiver Calibration

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***Abstract*—Periodic delay measurements provide fine range information in passive radar, while receiver calibration errors couple measurements collected from different illuminators. This paper develops a Bayesian localization method that accounts for frequency-dependent calibration, unequal ambiguity periods, and correlated measurement errors. Analytical integration of the calibration parameters reduces each receiver likelihood to a correlated Gaussian lattice sum. A conditional expansion bounds the unprocessed integer components, and observation-dependent bounds restrict the position search. Combining these bounds gives a stopping criterion that controls the total variation of the finite-grid posterior and the error in its mean. An analysis of source contrasts identifies the geometric information removed with the calibration parameters and explains how calibration priors change the relative weights of position ambiguities. The method is evaluated through 48600 factor computations, 27000 position updates, and finite-waveform simulations. At a posterior tolerance of 10^{-4}, the observation-dependent bound provides a median paired speedup of 1.75 over the coarse-range bound in the broad-error setting. Under the nominal measurement model, the position root-mean-square error is 14.87 m, compared with 20.92 m for calibration-invariant source differences. The waveform results connect localization accuracy to pilot resources, reference quality, and interference processing.**

***Index Terms*—Passive radar, Bayesian estimation, factor graphs, receiver calibration, integer ambiguity, numerical error bounds.**

I. Introduction

PASSIVE radar uses existing transmissions to observe a target from spatially separated receivers. The direct signal provides a reference for measuring the excess propagation delay of the target echo. Combining several illuminators and receivers improves geometric diversity, but also introduces nuisance parameters associated with the propagation paths and receiving channels. Recent studies have addressed localization with unknown transmitter positions [1], joint localization and sensor calibration [2], heterogeneous passive measurements [3], and direct-path interference [4]. These developments motivate inference methods that retain the dependence between target position and receiver errors.

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Code and selected data: https://github.com/windy07/error-controlled-passive-radar.

Factor graphs provide a convenient representation of this dependence [5], [6]. In distributed radar, message passing has been used to handle outliers [7], range deception [8], and direct position estimation [9]; related graphical models are developed in [10]. Expectation propagation [11] reduces the cost of continuous messages by matching moments, and recent passive-radar algorithms combine localization with target detection in the time or frequency domain [12], [13]. Joint estimation of target states and sensor registration parameters also appears in hierarchical tracking models [14] and multisensor association and bias compensation [15]. The accuracy of the resulting inference depends on both the measurement model and the approximation used to compute its messages.

Two features of multiband passive measurements are particularly relevant here. First, the residual response of a receiver varies with frequency. Measurements obtained through a common reference–surveillance chain pair therefore share calibration coefficients, even though their range biases differ across illuminators. Multiband calibration and distributed channel sounding provide practical examples of this structure [16], [17]. Second, a regularly spaced set of known frequency resources produces a periodic delay response. The period depends on the selected resources [18], while additional reference, control, or payload information can change the delay-estimation performance [19]. A likelihood that combines coarse and periodic measurements must consequently retain their correlations, the source-dependent periods, and the shared calibration parameters.

Integer ambiguity has a long history in satellite positioning. Integer least-squares methods exploit decorrelation to accelerate the search [20], while mixture filtering [21] and positional likelihood fields [22] retain several plausible ambiguities. Wrapped and partially wrapped distributions provide corresponding probabilistic models [23], [24]. For radar localization, integrating a shared Gaussian calibration vector produces a multivariate wrapped Gaussian whose components are coupled across sources. Selecting one joint component yields a max-mixture approximation, as in robust mapping [25] and max-log detection [26]. Retaining the sum instead requires evaluating the relative contributions of all relevant components. The distinction also affects

local curvature and uncertainty estimates for Gaussian mixtures [27].

Several established tools can evaluate or approximate these sums. Scalar wrapped densities admit efficient series evaluations [28], and periodic quadrature benefits from rapid convergence for analytic integrands [29]. Multidimensional Gaussian sums are related to Riemann theta functions [30]–[32]; lattice search offers another approach [33]. Error analysis for approximate messages [34] and bounds on partition functions [35] show the value of carrying numerical uncertainty through probabilistic inference. In a position search, however, a factor tolerance addresses only one part of the calculation. The posterior normalization also depends on positions at which the likelihood has yet to be evaluated. A useful stopping rule must account for both contributions.

This paper considers a stationary target observed by four silent receivers connected to a fusion processor. We develop marginal position inference under a frequency-dependent calibration model and a joint coarse–periodic measurement likelihood. The main contributions are as follows.

1) We derive receiver likelihoods that integrate the continuous calibration parameters and retain the joint integer ambiguities. A calibration-invariant contrast analysis identifies the geometric directions removed by source differencing, and a Gaussian decomposition shows how a proper calibration prior reweights competing ambiguities.
2) We construct positive bounds for conditional lattice expansions and combine them with observation-dependent position bounds. The resulting posterior interval gives explicit total-variation and position-mean error bounds on a prescribed grid. Rectangular spatial bounds extend the same calculation to dense grids.
3) We evaluate the method against complete-sphere and classical theta computations, calibration-invariant differences, and joint maximum components. Controlled measurement and finite-waveform results relate the computational savings, calibration benefit, and front-end sensitivity to the corresponding parts of the model.

Section II introduces the physical and statistical models. Section III derives the marginal likelihood and the statistical comparisons. Sections IV and V develop the computation and error analysis, respectively. Section VI presents the numerical results, followed by the conclusions in Section VII.

Notation: Vectors and matrices are distinguished by context. The symbols $\|\cdot\|$, $|\cdot|$, and $\succeq$ denote the Euclidean norm, determinant, and positive-semidefinite order, respectively; an absolute value applied to a vector is componentwise. The density of a zero-mean Gaussian vector with covariance $C \succ 0$ is ϕ_C. Expectations, covariance, total variation, and Kullback–Leibler divergence are denoted by $\mathbb{E}$, Cov, TV, and KL.

II. Signal and Measurement Models

A. Passive Geometry and Receiving Channels

Consider M stationary illuminators at known positions $t_m \in \mathbb{R}^2$ and four receivers at known positions $r_n \in \mathbb{R}^2$. A single target, whose presence is given, occupies an unknown position $x \in \Omega \subset \mathbb{R}^2$. The receivers operate passively and share their observations through wired connections. Each illuminator is available at every receiver in the nominal configuration. Relative to the direct path, the target echo has excess path length

$$h_{nm}(x) = \|x - t_m\| + \|x - r_n\| - \|t_m - r_n\|. \quad (1)$$

The coordinates and h_{nm} are in metres, and the corresponding excess delay is h_{nm}/c, where c is the propagation speed. Figure 1 shows the geometry used in the four-illuminator experiments.

The illuminators occupy separable frequency bands. Let $s_m(t)$ be the transmitted complex envelope and $\mathcal{H}_n^r$, $\mathcal{H}_n^s$ the reference and surveillance channel operators. A received stream is represented by

$$\begin{aligned} r_{nm}(t) &= \alpha_{nm}^r (\mathcal{H}_n^r s_m)(t - \tau_{nm}^r) + w_{nm}^r(t), \\ v_{nm}(t) &= \alpha_{nm}^t (\mathcal{H}_n^s s_m)(t - \tau_{nm}^r - h_{nm}/c) e^{\mathrm{i}2\pi\nu_{nm}t} \\ &\quad + \alpha_{nm}^d (\mathcal{H}_n^s s_m)(t - \tau_{nm}^r) + w_{nm}^s(t). \end{aligned} \quad (2)$$

Here α_{nm}^r, α_{nm}^t, and α_{nm}^d are unknown complex gains; τ_{nm}^r contains the direct propagation delay and capture epoch. The noises are denoted by w_{nm}^r and w_{nm}^s. The nominal stationary scene has $\nu_{nm} = 0$; a nonzero value represents the residual frequency offset examined later. The waveform experiments use scalar, ideally channelized streams, with received signal-to-noise ratio (SNR) specifying the signal and noise powers.

A shared sampling clock removes the common timing origin when the reference and echo delays are differenced. Their differential transfer response remains. With $H_n^r(f)$ and $H_n^s(f)$ denoting the channel frequency responses, its group delay in range units is

$$b_n(f) = -\frac{c}{2\pi}\frac{d}{df}\arg\left(\frac{H_n^s(f)}{H_n^r(f)}\right). \quad (3)$$

For a smooth response over the processed bands, write

$$\begin{aligned} b_n(f_m) &= a_m^T \beta_n + e_{nm}, \\ a_m &= \begin{bmatrix} 1 & (f_m - f_*)/F_* \end{bmatrix}^T. \end{aligned} \quad (4)$$

The frequencies f_m, f_*, and F_* are in hertz. Thus a_m is dimensionless and the two coefficients in β_n are in metres. More generally, $A = [a_1, \ldots, a_M]^T \in \mathbb{R}^{M\times q}$ contains a prescribed full-rank basis. The term

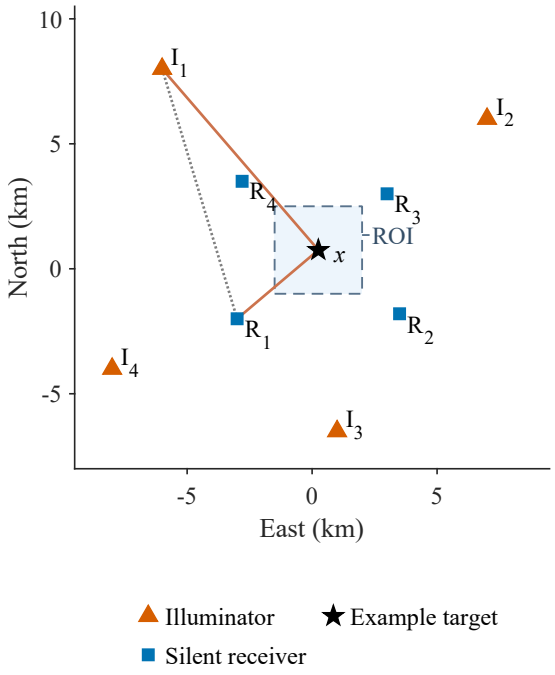


Fig. 1. Geometry of the four-receiver network. The highlighted bistatic path and its direct-path reference define the excess range in (1). The shaded region specifies the position support for the measurement-level study; the target symbol marks an illustrative position within it.

e_{nm} represents the remaining frequency dependence. A common coefficient vector is appropriate for bands that traverse the same channel pair; separate receiver chains are represented by separate coefficient blocks. Equation (4) follows the frequency-response description used in multiband calibration [16] and is tested below through an additional quadratic term.

The fine-delay period is determined by the frequency resources used in its estimator. For known tones at $f_0 + kK_m\Delta f_m$ and an unknown common complex gain, increasing the delay by $1/(K_m\Delta f_m)$ preserves the fitted response. In range units,

$$L_m = \frac{c}{K_m \Delta f_m}, \qquad D = \mathrm{diag}(L_1, \ldots, L_M). \tag{5}$$

Coherent processing of several comb offsets can extend this period. We use the period of the actual resource set in each likelihood, including the distinction between one-symbol and combined-symbol pilot processing [18].

B. Correlated Coarse and Periodic Measurements

At receiver n, let $z_{0n} \in \mathbb{R}^M$ collect the coarse ranges and let $y_n \in \prod_{m=1}^{M}[0, L_m)$ collect the fine ranges reduced modulo their periods. Define $h_n(x) = [h_{n1}(x), \ldots, h_{nM}(x)]^T$, $H = [I_M; I_M]$, and $B = HA$. The observation model is

$$\begin{bmatrix} z_{0n} \\ y_n + Dk_n \end{bmatrix} = Hh_n(x) + B\beta_n + \varepsilon_n, \tag{6}$$

where $k_n \in \mathbb{Z}^M$ contains the unknown integer ambiguities and $\varepsilon_n \sim \mathcal{N}(0, R_n)$. The covariance $R_n \succ 0$, in squared metres, includes correlations between sources and between the coarse and fine measurements. The shared calibration contribution is represented explicitly by $B\beta_n$.

Calibration information available before the current observation is described by

$$\beta_n \sim \mathcal{N}(\mu_n, \Sigma_n), \qquad x \sim \pi(x), \tag{7}$$

with $\Sigma_n \succ 0$ on the uncertain coefficient subspace and a proper position density π. The model parameters are fixed during an update. We assume independent receiver errors and calibration priors conditional on the modeled quantities. This assumption gives the factorization

$$\begin{aligned} &p(x, \beta, k \mid z_0, y) \\ &\quad \propto \pi(x) \prod_{n=1}^{4} \phi_{\Sigma_n}(\beta_n - \mu_n)\psi_n(x, \beta_n, k_n), \\ &\psi_n = \phi_{R_n}\left(\begin{bmatrix} z_{0n} \\ y_n + Dk_n \end{bmatrix} - Hh_n(x) - B\beta_n\right). \end{aligned} \tag{8}$$

Correlated observations from the same receiver enter one factor, as illustrated in Fig. 2. A shared uncertainty across receivers would enter a joint factor or an additional common variable. For sequential operation, π is replaced by the predictive distribution, with any position–calibration dependence carried in the joint prior. The present study evaluates individual position updates with independently supplied priors.

III. Marginal Position Inference

A. Elimination of Receiver Calibration

The linear dependence on β_n permits analytical integration before the position search. To simplify the derivation, suppress the receiver index and partition

$$C = R + B\Sigma B^T = \begin{bmatrix} C_{00} & C_{01} \\ C_{10} & C_{11} \end{bmatrix}. \tag{9}$$

The integrated coarse residual and the conditional fine distribution are specified by

$$\begin{aligned} e_0(x) &= z_0 - h(x) - A\mu, \\ G &= C_{10}C_{00}^{-1}, \qquad S = C_{11} - C_{10}C_{00}^{-1}C_{01}, \\ m_1(x) &= h(x) + A\mu + Ge_0(x). \end{aligned} \tag{10}$$

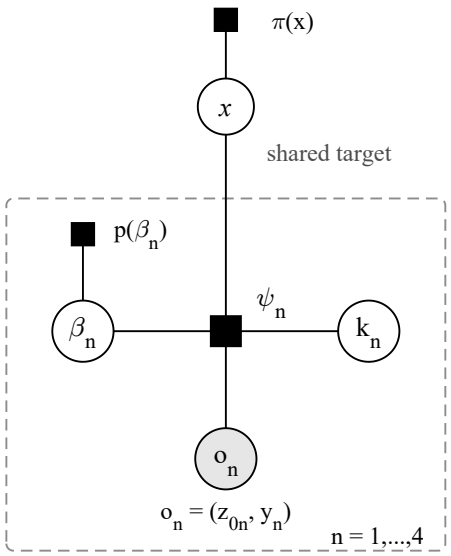


Fig. 2. Receiver-wise factorization of the joint distribution. The calibration vector and integer ambiguities connect the source measurements within each receiver factor. Eliminating these variables produces a likelihood message to the shared target position.

Completing the Gaussian square, followed by summation over the possible integers, gives

$$F_n(x) = \phi_{C_{00}}\{e_0(x)\}\, W_{S,D}\{y_n - m_1(x)\}, \quad (11)$$

$$W_{S,D}(u) = \sum_{k\in\mathbb{Z}^M} \phi_S(u + Dk). \quad (12)$$

Equation (11) separates the coarse-data density from the fine-data density conditioned on the coarse measurement. In particular, e_0 shifts the conditional fine mean through G, while the Schur complement S specifies the remaining uncertainty.

The position posterior and its minimum mean-square-error estimate are

$$p(x \mid z_0, y) = \frac{\pi(x)\prod_n F_n(x)}{\int_\Omega \pi(\xi)\prod_n F_n(\xi)\, d\xi}, \quad (13)$$

$$\widehat{x} = \int_\Omega x\, p(x \mid z_0, y)\, dx. \quad (14)$$

Elimination reduces the factor graph to receiver messages and a normalization at the position variable. The main computational task is therefore the evaluation of (12) over the positions contributing to (13). Section IV develops a finite-grid implementation with a prescribed posterior error.

B. Calibration-Invariant Source Differences

Source differences provide a useful comparison because they remove shared calibration without requiring its probability distribution. Their covariance must reflect the common observations used in forming the differences. Let $R = L_R L_R^T$ and choose an orthogonal matrix $Q = [Q_1, Q_2]$ such that $\mathrm{col}(Q_1) = \mathrm{col}(L_R^{-1}B)$. The transformed residual $Q_2^T L_R^{-1} v_k$, where

$$v_k = \begin{bmatrix} z_0 \\ y + Dk \end{bmatrix} - Hh(x), \quad (15)$$

is invariant to β. Its quadratic form uses the projected precision

$$P = R^{-1} - R^{-1}B(B^T R^{-1} B)^{-1} B^T R^{-1}. \quad (16)$$

This is the weighted elimination of a linear nuisance parameter, closely related to variable projection [36]. Integrating β with respect to flat measure gives the same position-dependent quadratic form, with a constant Gaussian normalization.

Set $v_0 = [z_0 - h(x); y - h(x)]$ and $E = [0; D]$, and define

$$V_d = (E^T P E)^{-1}, \qquad u_d = V_d E^T P v_0,$$
$$q_d = v_0^T P v_0 - u_d^T V_d^{-1} u_d. \quad (17)$$

The required inverse exists: $Ek \in \mathrm{col}(B)$ implies $Ek = Bb$, whose coarse block gives $Ab = 0$; full column rank of A then gives $b = 0$ and $k = 0$. Completing the square yields

$$\sum_{k\in\mathbb{Z}^M} e^{-v_k^T P v_k/2} = e^{-q_d/2}(2\pi)^{M/2}|V_d|^{1/2} W_{V_d,I}(u_d). \quad (18)$$

We use this generalized source-difference likelihood as the calibration-invariant baseline. It retains the joint integer sum and the full covariance of the contrasts. Keeping both Q_1 and Q_2 blocks, together with the original prior, instead recovers (11) through an invertible change of coordinates.

The loss of a common propagation leg under differencing has been discussed for passive localization in [37, App. A.2]. The frequency basis makes its consequence for this network explicit.

Proposition 1. *Assume that* $\mathbf{1} \in \mathrm{col}(A)$ *and that every receiver has an unconstrained coefficient vector with the same basis* $A \in \mathbb{R}^{M\times q}$*. The position Jacobian of calibration-invariant source contrasts has rank at most* $\min\{2, M - q\}$*, for any number of receivers.*

Proof. Choose $N \in \mathbb{R}^{M\times(M-q)}$ with $N^T A = 0$ and full column rank. Decompose the range vector as

$$h_n(x) = g(x) + \mathbf{1}\|x - r_n\| - d_n, \quad (19)$$

where $g_m(x) = \|x - t_m\|$ and $d_{nm} = \|t_m - r_n\|$. Since $N^T\mathbf{1} = 0$, differentiation gives

$$\frac{\partial(N^T h_n)}{\partial x} = N^T \frac{\partial g}{\partial x}. \quad (20)$$

The right-hand side is common to all receivers and has $M-q$ rows and two columns. Stacking receiver contrasts preserves this rank bound. □

For the affine basis in (4), local two-dimensional identification through these contrasts requires at least four illuminators and suitable source geometry. Additional receivers improve precision along the retained directions. A finite calibration covariance also provides information in the directions removed by the contrasts.

C. Calibration Priors and Integer Ambiguities

The influence of calibration information can be identified for each integer component. Define

$$\Lambda = B^T R^{-1} B, \qquad \widehat{\beta}_k = \Lambda^{-1} B^T R^{-1} v_k, \tag{21}$$

and let $g_k(x) = \int \phi_R(v_k - B\beta)\, d\beta$ be its flat-calibration likelihood. The quadratic decomposition underlying (16) gives

$$\phi_R(v_k - B\beta) = g_k(x)\, \phi_{\Lambda^{-1}}(\beta - \widehat{\beta}_k). \tag{22}$$

Multiplication by the calibration prior and integration now produce

$$\phi_C(v_k - B\mu) = g_k(x)\, \phi_{\Sigma+\Lambda^{-1}}(\widehat{\beta}_k - \mu), \tag{23}$$

$$F(x) = G_{\mathrm{flat}}(x) \sum_k \pi_k^{\mathrm{flat}}(x) \phi_{\Sigma+\Lambda^{-1}}(\widehat{\beta}_k - \mu), \tag{24}$$

where $G_{\mathrm{flat}} = \sum_k g_k$ and $\pi_k^{\mathrm{flat}} = g_k / G_{\mathrm{flat}}$. Both sums converge because $E^T P E \succ 0$. Equation (23) shows that an ambiguity receives additional weight according to the calibration displacement it requires. A geometrically plausible alias can therefore receive little posterior mass when its fitted coefficients lie far from the calibration prior. This mechanism will explain the error-tail comparison in Section VI-B.

To assess the effect of retaining several ambiguities, we also replace $W_{S,D}(u)$ by $\max_k \phi_S(u + Dk)$. The integer maximum is computed jointly by closest-lattice-point sphere search, preserving the covariance and calibration prior. Its position objective is normalized and its mean is reported, so both estimates use the same position loss. This comparison is the joint maximum-component approximation associated with max-mixture and max-log methods [25], [26]. When one component carries most of the likelihood, the sum and maximum produce similar estimates. The computation study separately compares alternative evaluators of the full sum.

IV. Bounded Posterior Computation

The position integral contains two potentially large searches: the integer sum at each receiver and the set of candidate positions. This section assigns upper bounds to the unresolved mass in both searches. Their sum determines when the requested posterior accuracy has been reached.

A. Conditional Expansion of the Lattice Likelihood

Normalize the measurement periods using $V = D^{-1} S D^{-1}$ and $t = D^{-1} u$. Then

$$W_{S,D}(u) = |D|^{-1} \sum_{k \in \mathbb{Z}^M} \phi_V(t + k). \tag{25}$$

For $V = LL^T$, the Gaussian chain rule expresses every term as

$$\phi_V(t + k) = \prod_{j=1}^{M} \phi_{d_j^2}\{k_j - \mu_j(k_{1:j-1})\}, \tag{26}$$

where $d_j = L_{jj}$ and

$$\mu_j = -t_j + L_{j,1:j-1} L_{1:j-1,1:j-1}^{-1} (t_{1:j-1} + k_{1:j-1}). \tag{27}$$

The second term is zero for $j = 1$. A partial integer vector therefore specifies a branch of a conditional search tree.

The unexpanded descendants can be bounded without enumerating their integers. Poisson summation gives the scalar identity

$$\sum_{k \in \mathbb{Z}} \phi_{d^2}(k - \mu) = 1 + 2 \sum_{\ell=1}^{\infty} e^{-2\pi^2 d^2 \ell^2} \cos(2\pi \ell \mu) \le \sum_{k \in \mathbb{Z}} \phi_{d^2}(k) \equiv B(d). \tag{28}$$

The positivity of the Fourier coefficients places the maximum at an integer center, consistently with scalar wrapped-normal evaluations [28]. A prefix with accumulated weight w and next coordinate j consequently has total descendant mass at most $w \prod_{\ell=j}^{M} B(d_\ell)$. For a remaining half-line whose closest integer is at distance $b > 0$, monotonicity of the Gaussian density also gives

$$\sum_{v=0}^{\infty} \phi_{d^2}(b + v) \le \phi_{d^2}(b) + \int_b^{\infty} \phi_{d^2}(z)\, dz = \phi_{d^2}(b) + \frac{1}{2} \operatorname{erfc}\left(\frac{b}{\sqrt{2} d} \right). \tag{29}$$

An expansion records the selected integers and bounds both remaining half-lines. In this way the frontier continues to cover the entire integer lattice.

Theorem 2. *Let the evaluated leaves and frontier branches form a disjoint partition of $\mathbb{Z}^M$. Denote the accumulated leaf density, including $|D|^{-1}$, by Z_L, and the sum of valid upper bounds on the frontier density by U. Then*

$$W_{S,D}(u) \in [Z_L, Z_L + U]. \tag{30}$$

Proof. Equation (26) assigns a nonnegative density to every leaf. Repeated application of (28) bounds the descendants of a fixed prefix, while (29) accounts for its unenumerated integer tails. Summing these bounds over disjoint frontier branches gives an upper bound on the mass outside the evaluated leaves. An expansion replaces one branch by a disjoint collection of children and tails, preserving the argument at the next iteration. □

The implementation expands the frontier entry with the largest bound and accepts a factor when $U/Z_L \le \epsilon$. At a resource limit it returns the current interval and completion status. With the direct frontier array used here, T candidate visits require up to $O(T^2 + TM)$ operations and $O(TM)$ storage. Bounds in this section concern truncation in real arithmetic; the MATLAB implementation evaluates logarithmic weights in double precision, and numerical comparisons check the resulting densities and estimates.

B. Posterior Normalization and Position Selection

Let $\mathcal{G} = \{x_i\}_{i=1}^P$ be a fixed position grid with prior weights π_i. The exact unnormalized and normalized weights are

$$w_i = \pi_i \prod_{n=1}^{4} F_n(x_i), \qquad p_i = \frac{w_i}{\sum_j w_j}. \tag{31}$$

Suppose $l_i \le w_i \le u_i$ at each point, with $l_i = 0$ at unevaluated positions. Write

$$L = \sum_i l_i > 0, \qquad \Delta = \sum_i (u_i - l_i), \qquad q_i = l_i/L. \tag{32}$$

Using $\mathrm{TV}(p,q) = \frac{1}{2}\sum_i |p_i - q_i|$, the following result converts these intervals into a posterior stopping criterion.

Theorem 3. *For the weights in* (31)–(32)*, let D_Ω be an upper bound on the diameter of $\mathcal{G}$. The normalized approximation q satisfies*

$$\mathrm{TV}(p,q) \le \eta \equiv \frac{\Delta}{L+\Delta}, \tag{33}$$

$$\left\| \sum_i x_i p_i - \sum_i x_i q_i \right\| \le D_\Omega \eta. \tag{34}$$

Proof. Set $\delta_i = w_i - l_i \ge 0$ and $\delta = \sum_i \delta_i \le \Delta$. If $\delta > 0$, define $r_i = \delta_i/\delta$ and express the exact posterior as

$$p = \frac{L}{L+\delta} q + \frac{\delta}{L+\delta} r. \tag{35}$$

Consequently,

$$\mathrm{TV}(p,q) = \frac{\delta}{L+\delta}\,\mathrm{TV}(r,q) \le \frac{\delta}{L+\delta} \le \frac{\Delta}{L+\Delta}. \tag{36}$$

The two means of r and q lie in the convex hull of the grid and are separated by at most D_Ω. Taking means in the mixture identity proves (34). For $\delta = 0$, $p = q$ and both bounds follow directly. □

Factor intervals give bounds at evaluated positions by multiplication of their nonnegative endpoints. For the remaining positions, the first conditional factor gives the inexpensive envelope

$$W_{S,D}(u) \le |D|^{-1} W_{d_1^2,1}(t_1) \prod_{j=2}^{M} B(d_j). \tag{37}$$

This retains the observed value of the first normalized residual. Replacing it by $B(d_1)$ gives the coarse-range envelope used for comparison. Multiplication by the coarse densities and the prior produces a position bound b_i. Ordering positions by b_i places likely contributions early in the search, while the cumulative envelope bounds the unvisited remainder.

Algorithm 1 allocates one posterior budget to factor truncation and skipped positions. The envelope controls evaluation order and omitted mass; the evaluated likelihood remains (11). The result is compatible with any lattice evaluator that returns valid intervals. This allows the numerical benefit of the envelope to be measured independently of the chosen factor evaluator.

C. Reference Lattice Evaluators

For a direct comparison with established Gaussian-sum methods, define the Riemann theta function by $\Theta(z \mid \Xi) = \sum_k \exp(\pi \mathrm{i} k^T \Xi k + 2\pi \mathrm{i} k^T z)$. The normalized factor has the representation

$$W_{V,I}(t) = \phi_V(t)\,\Theta\left(\frac{\mathrm{i}V^{-1}t}{2\pi} \,\middle|\, \frac{\mathrm{i}V^{-1}}{2\pi}\right). \tag{38}$$

We implement a classical theta reference using Siegel transformations [31] and the ellipsoid remainder of [30]. For imaginary part Y after transformation and shortest nonzero distance ρ in $\sqrt{\pi}\,\mathrm{chol}(Y)\mathbb{Z}^M$, the scaled remainder obeys

$$E_R \le \frac{M}{2}\left(\frac{2}{\rho}\right)^M \Gamma\left(\frac{M}{2}, (R - \rho/2)^2\right), \tag{39}$$

for $R > \sqrt{M/2} + \rho/2$. Here $\Gamma(a,z)$ is the upper incomplete gamma function. Exact shortest-vector enumeration supplies ρ, and the absolute remainder gives an interval for the transformed density. Covariance-dependent preparation is included in the timing comparison.

A second reference enumerates a complete Mahalanobis sphere, with its radius selected from a positive

Algorithm 1 Marginal position update with bounded posterior error

Require: Receiver observations and model parameters; grid (x_i, π_i); tolerances ϵ, η_*; search budgets.

Form the conditional residuals and covariances in (10).
Compute b_i from (37); sort positions by decreasing b_i and accumulate the remaining bounds.
Set $\mathcal{V} = \varnothing$ and $L = 0$.
while position evaluations remain within budget **do**
    Select the next batch in the sorted order.
    Evaluate each receiver factor in the batch, retaining its interval at tolerance ϵ or at the lattice budget.
    Form $[l_i, u_i]$ and add the batch to $\mathcal{V}$.
    $L \leftarrow \sum_{i\in\mathcal{V}} l_i$.
    $\Delta \leftarrow \sum_{i\in\mathcal{V}}(u_i - l_i) + \sum_{i\notin\mathcal{V}} b_i$.
    if $L > 0$ and $\Delta/(L+\Delta) \le \eta_*$ **then**
        Terminate the search.
    end if
end while
For $L > 0$, return $q_i = l_i/L$, its moments, η, and completion status; otherwise return the unresolved-mass status.

Babai term and a Gaussian Chernoff bound on the remaining lattice mass. A vectorized integer-box reference uses $D_V = \mathrm{diag}(\sum_j |V_{ij}|) \succeq V$ and

$$\phi_V(v) \le \sqrt{|D_V|/|V|}\, \phi_{D_V}(v) \tag{40}$$

to bound its omitted terms through scalar tails. For diagonal covariance the sum factors into scalar wrapped densities, giving a specialized reference. All these evaluators target the same likelihood in (12).

D. Spatial Bounds for Dense Grids

A dense uniform grid can be searched through rectangular blocks before its individual points are evaluated. Let a block have center x_C, half-widths w_C, and radius $r_C = \|w_C\|$. For a fixed vector λ, the geometry in (1) gives

$$|\lambda^T\{h(x) - h(x_C)\}| \le r_C \left(\sum_m |\lambda_m| + \left|\sum_m \lambda_m\right|\right). \tag{41}$$

Each source distance is 1-Lipschitz, and the common receiver distance is multiplied by $\sum_m \lambda_m$. This form retains the cancellation of the receiver leg in source contrasts. On blocks separated from the sensors, a distance-Hessian bound supplies a tighter Taylor remainder.

Apply (41) to the rows of the whitened map $V^{-1/2}D^{-1}(I - G)$. For $t(x) = D^{-1}\{y - m_1(x)\}$, this gives a computable ρ_C satisfying

$$\|V^{-1/2}\{t(x) - t(x_C)\}\| \le \rho_C. \tag{42}$$

For any $\gamma > 1$, completing a square establishes

$$\|z + v\|^2 \ge \gamma^{-1}\|z\|^2 - (\gamma - 1)^{-1}\|v\|^2. \tag{43}$$

Using this inequality for every Gaussian component, with its normalization retained, yields

$$W_{V,I}\{t(x)\} \le \gamma^{M/2} e^{\rho_C^2/[2(\gamma-1)]} W_{\gamma V,I}\{t(x_C)\}. \tag{44}$$

Multiplication by $|D|^{-1}$ restores physical units. The bound retains correlation as a block contracts; at zero radius the center is evaluated directly.

The common position displacement can also be used across receivers. Select one legal integer component at each receiver and write its whitened residuals, together with the coarse residuals, as $v_j - G_j d - e_j(d)$, where $d = x - x_C$ and $\|e_j(d)\| \le \rho_j$. Expanding the square gives

$$-\tfrac{1}{2}\|v_j - G_j d - e_j\|^2 \le -\tfrac{1}{2}\|v_j\|^2 + v_j^T G_j d + \|v_j\|\rho_j. \tag{45}$$

If ℓ_C is the selected joint log density at the center, set $g_C = \sum_j G_j^T v_j$ and $\epsilon_C = \sum_j \|v_j\|\rho_j$. The selected product is bounded by

$$U_\star = \exp\{\ell_C + |g_C|^T w_C + \epsilon_C\}. \tag{46}$$

Summing the gradients before maximizing retains the requirement that all receivers share the same displacement.

Let a_n^+ bound the selected fine component, b_n^+ bound its remaining aliases, and Q_C bound the coarse-factor product. All other joint components are included by the positive recursion

$$\begin{aligned} P_n &= P_{n-1}a_n^+, \\ E_n &= E_{n-1}(a_n^+ + b_n^+) + P_{n-1}b_n^+, \\ U_C^{\mathrm{joint}} &= U_\star + Q_C E_4, \end{aligned} \tag{47}$$

initialized with $P_0 = 1$ and $E_0 = 0$. The remaining-alias bound uses (44) and the positive lattice remainder; a full-factor envelope is available when that calculation reaches its budget. Taking the smaller of the joint and receiver-wise bounds preserves validity.

For uniform midpoint weights, a block containing n_C points has unnormalized mass at most $n_C U_C/P$. The algorithm bisects the block with the largest mass bound and evaluates small blocks directly. A preliminary coarse-grid estimate guides initial visits, with its cost included in the update. The accumulated block remainders enter Δ in (32), so the same posterior stopping rule applies throughout the spatial search.

V. Information and Estimation Error

A. Information in the Marginal Measurements

The marginal model permits an information analysis that includes uncertainty in both calibration and integer ambiguity. Let $J_h = \partial h/\partial x \in \mathbb{R}^{M\times 2}$ and $T = I - G$.

At a fine residual u, define the component probabilities and conditional residual mean

$$\omega_k(u) = \frac{\phi_S(u + Dk)}{W_{S,D}(u)}, \qquad \bar{r}(u) = \sum_k \omega_k(u)(u + Dk). \tag{48}$$

Gaussian decay ensures uniform convergence of the series and its derivatives on compact sets. Differentiating (11) therefore gives the receiver score

$$s_x = J_h^T\{C_{00}^{-1}e_0 + T^T S^{-1}\bar{r}(y - m_1)\}. \tag{49}$$

The conditional fine score has mean zero given z_0. Moreover, its information is invariant to translation of the periodic domain. The cross term in $\mathbb{E}[s_x s_x^T]$ consequently vanishes, leaving

$$J_x = J_h^T\{C_{00}^{-1} + T^T J_W T\}J_h. \tag{50}$$

To evaluate J_W, let $r \sim \mathcal{N}(0, S)$ and denote reduction modulo D by $[r]_D$. The conditional-score identity and the law of total covariance give

$$\begin{aligned} J_W &= S^{-1}\,\mathrm{Cov}\{\mathbb{E}[r \mid [r]_D]\}S^{-1} \\ &= S^{-1} - S^{-1}\mathbb{E}\{\mathrm{Cov}(r \mid [r]_D)\}S^{-1}. \end{aligned} \tag{51}$$

This is the expected-information form of the missing-information identity [38]. The second term quantifies the information removed by unresolved integer ambiguities. It approaches zero as the conditional integer distribution concentrates on one component.

Calibration has a corresponding effect in the unwrapped model. Woodbury's identity yields

$$\begin{aligned} H^T C^{-1} H = H^T R^{-1} H - H^T R^{-1} B \\ \times (\Sigma^{-1} + B^T R^{-1} B)^{-1} B^T R^{-1} H. \end{aligned} \tag{52}$$

The calibration precision Σ^{-1} reduces the subtraction associated with the nuisance subspace. Taking the flat-calibration limit removes this precision and recovers the projected expression in (16). Equations (51) and (52) thus separate ambiguity uncertainty from calibration uncertainty within the same measurement model.

For a prior-averaged position comparison, we use the vector Van Trees inequality [39]. On the rectangle $|x_j - c_j| \le w_j/2$, choose

$$\pi(x) = \prod_{j=1}^{2} \frac{2}{w_j}\cos^2\!\left(\frac{\pi(x_j - c_j)}{w_j}\right), \qquad J_\pi = \mathrm{diag}\!\left(\frac{4\pi^2}{w_1^2}, \frac{4\pi^2}{w_2^2}\right). \tag{53}$$

The density vanishes on the boundary. Under the usual differentiability and finite-information conditions, the joint score s satisfies $\mathbb{E}[(\widehat{x} - X)s^T] = I$ by integration by parts. With

$$\mathcal{J} = \mathbb{E}_\pi\left[\sum_n J_{x,n}(X)\right] + J_\pi, \tag{54}$$

positive semidefiniteness of the second moment of $(\widehat{x} - X) - \mathcal{J}^{-1}s$ gives

$$\mathbb{E}[(\widehat{x} - X)(\widehat{x} - X)^T] \succeq \mathcal{J}^{-1}. \tag{55}$$

This prior-averaged bound applies to the posterior-mean estimator used in the controlled measurement study. The waveform study instead uses an externally specified uniform region and paired empirical comparisons.

B. Model, Grid, and Numerical Contributions

The value of a fine measurement can first be stated under a common correctly specified probability law. Set $\mu_0 = \mathbb{E}[X \mid Z_0]$ and $\mu_{01} = \mathbb{E}[X \mid Z_0, Y]$. Orthogonality of conditional expectations gives

$$\mathbb{E}\|X - \mu_0\|^2 - \mathbb{E}\|X - \mu_{01}\|^2 = \mathbb{E}\|\mu_{01} - \mu_0\|^2. \tag{56}$$

Hence the additional observation reduces Bayes risk by the mean squared change it induces in the optimal estimate. Comparing this identity with finite-waveform results helps identify the effect of the compressed measurement law.

For a quantitative decomposition, let P be the data-generating law of (X, O), $O = (Z_0, Y)$, and let Q be the inference model, with the same prior on a region of diameter D_Ω. Assume $K = \mathrm{KL}(P_{X,O}\|Q_{X,O}) < \infty$. The KL chain rule and Pinsker's inequality [40] give

$$\mathbb{E}_{P_O}\,\mathrm{TV}(P_{X|O}, Q_{X|O})^2 \le K/2. \tag{57}$$

Let μ_P and μ_Q be the corresponding continuous posterior means. Let μ_h be the exact mean on the chosen grid and $\widetilde{\mu}_h$ its numerical approximation. Define the grid error $G_h = \|\mu_h - \mu_Q\|_{L^2(P_O)}$ and $\eta_2 = (\mathbb{E}_{P_O}\eta_h^2)^{1/2}$, where η_h is the achieved finite-grid TV bound. Inserting the two intermediate means and applying Minkowski's inequality gives

$$\begin{aligned} &\|\widetilde{\mu}_h - \mu_P\|_{L^2(P_O)} \\ &\quad \le \|\mu_Q - \mu_P\|_{L^2(P_O)} + G_h + \|\widetilde{\mu}_h - \mu_h\|_{L^2(P_O)} \\ &\quad \le D_\Omega\sqrt{K/2} + G_h + D_\Omega\eta_2. \end{aligned} \tag{58}$$

Finally, conditional-mean orthogonality converts the mean error into excess position risk:

$$\begin{aligned} &\mathbb{E}_P\|X - \widetilde{\mu}_h\|^2 - \mathbb{E}_P\|X - \mu_P\|^2 \\ &\quad = \|\widetilde{\mu}_h - \mu_P\|_{L^2(P_O)}^2 \\ &\quad \le \min\left\{D_\Omega^2, \left(D_\Omega\sqrt{K/2} + G_h + D_\Omega\eta_2\right)^2\right\}. \end{aligned} \tag{59}$$

The three terms specify the roles of model characterization, spatial resolution, and numerical summation. Theorem 3 controls the last term on a fixed grid. Paired grid refinements assess the change in estimates as spatial resolution increases. Their root-mean-square change $\widehat{S}$ also bounds the absolute difference between the two empirical localization RMSEs by the reverse triangle inequality.

For the curvature and correlation perturbations used below, Gaussian pre-wrapping KL calculations give 5.9996 and 7.8667, respectively. Through data processing these are valid upper bounds for the wrapped observations, but (59) then reaches its diameter ceiling. In these settings, the controlled localization and coverage results provide the useful quantitative description of model sensitivity.

VI. Numerical Results

The evaluation addresses three questions: how much computation the bounds save, how calibration information affects localization, and how finite-waveform processing affects the measurement model. Competing methods use the same saved observations within each trial. Confidence intervals are 95% intervals obtained by resampling independent trials; links within a receiver remain together. Computation times are measured with single-thread MATLAB R2024b on an AMD Ryzen 7 6800H and include preparation, sorting, and likelihood evaluation.

A. Lattice and Position Computation

1) Factor evaluation: We first compare the conditional tree, complete-sphere, and classical theta evaluators with normalized covariance

$$V = s^2\{(1-\rho)I + \rho\mathbf{1}\mathbf{1}^T\}. \tag{60}$$

The settings $M \in \{4, 8, 12\}$, $s \in \{0.03, 0.15, 0.35\}$, and $\rho \in \{0, 0.8, 0.98\}$ give 27 covariance configurations. Each uses 100 inputs uniform on the fundamental torus and 100 wrapped Gaussian inputs, representing search residuals and model-generated residuals, respectively. Physical periods $L_j = 400 + 137j$ m check the scaling in (25). Three relative tolerances, 10^{-3}, 10^{-6}, and 10^{-9}, give 48600 factor calls. The candidate budget is 4000, and method order is randomized in batches of 50.

Table I shows that the conditional expansion completes more requests as dimension increases: at $M = 12$, it completes 4742 calls, compared with 4475 for the sphere and 3900 for theta. The cost of a completed call also depends on covariance structure. At $M = 12$, $s = 0.35$, $\rho = 0.98$, and tolerance 10^{-6}, theta is faster than sphere, with median paired time ratios 0.569 and 0.643 for the two input laws. Their intervals are $[0.502, 0.607]$ and $[0.609, 0.680]$, based on 99 and 100 common completions. For broad diagonal factors, all three general searches reach the candidate limit, while the scalar-product evaluator exploits the exact factorization. These results identify complementary regimes for the evaluators.

2) Position updates: The four-source geometry in Fig. 1 uses illuminator coordinates $(-6, 8)$, $(7, 6)$, $(1, -6.5)$, and $(-8, -4)$ km, and receiver coordinates $(-3, -2)$, $(3.5, -1.8)$, $(3, 3)$, and $(-2.8, 3.5)$ km. Positions are drawn uniformly from $[-1500, 2000] \times [-1000, 2500]$ m. Grid spacings 50, 25, 12.5, and 6.25 m give 4900, 19600, 78400, and 313600 midpoint samples. The nominal coarse/fine standard deviations are 150/30 m, and the calibration constant/slope standard deviations are 60/25 m. The measurement covariance is

$$R = \operatorname{diag}(a)\left(\begin{bmatrix} 1 & 0.2 \\ 0.2 & 1 \end{bmatrix} \otimes T_\rho\right)\operatorname{diag}(a), \tag{61}$$

where $a = [\sigma_0\mathbf{1}^T, \sigma_1\mathbf{1}^T]^T$ and $(T_\rho)_{ij} = 0.35^{|i-j|}$. A broad-error setting increases the measurement scales to 600/150 m and the calibration scales to 300/100 m.

The study includes five source counts, $M = 3, 4, 6, 8, 12$, at 25-m spacing, the three additional four-source grids, and the broad-error setting. Each of these nine configurations uses 200 trials. The source-count configurations use the first M coordinates in the saved geometry, with normalized frequencies equally spaced on $[-1, 1]$. Their periods follow (5) with 30-kHz spacing and the comb sequence $(2, 4, 6, 12, 4, 6, 12, 2, 4, 6, 12, 2)$. The four grid resolutions reuse identical observations.

Five implementations separate the position bound from the factor evaluator: sphere/coarse, sphere/prefix, tree/prefix, theta/coarse, and theta/prefix. Here "prefix" denotes (37). Each is evaluated at posterior TV tolerances 10^{-2}, 10^{-4}, and 10^{-6}, with factor tolerance 10^{-10} and a 20000-candidate limit. All 27000 position updates attain their requested posterior tolerance.

Figure 3 shows the increase in computation time with grid size. At 313600 points, replacing the coarse bound by the prefix bound gives a median paired speedup of 1.404 for the sphere evaluator, with interval $[1.382, 1.419]$. With the prefix bound fixed, the median

TABLE I
Factor evaluations reaching the requested tolerance. Each entry has 5400 attempts across covariance settings, input distributions, and tolerances.

Illuminators	Conditional tree	Sphere	Theta
4	5400	5400	5400
8	5000	4838	4792
12	4742	4475	3900

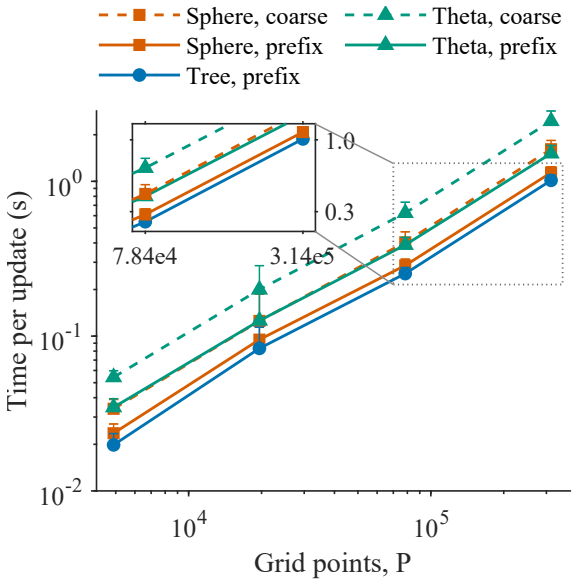


Fig. 3. Computation time as the four-source position grid is refined, at posterior TV tolerance 10^{-4}. Medians and upper 90th-percentile whiskers use 200 paired trials. The enlarged region separates the tree and sphere implementations using the same observation-dependent bound.

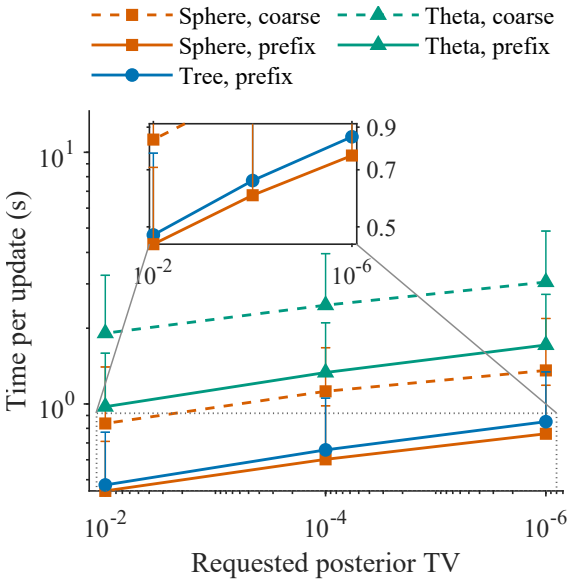


Fig. 4. Effect of posterior tolerance in the broad-error configuration. The observation-dependent envelope lowers the time for both sphere and theta evaluation. The inset resolves the positive-sum methods, for which sphere is faster in this setting. All 200 trials per point satisfy the requested tolerance.

times for tree, sphere, and theta are 1.014, 1.139, and 1.513 s. Thus the conditional tree uses 33.0% less median time than theta in this configuration. The paired sphere/tree ratio is 1.123, with interval $[1.115, 1.127]$.

The broad-error results in Fig. 4 further isolate the envelope contribution. At TV 10^{-4}, replacing the coarse bound by (37) gives median paired speedups of 1.751 for sphere and 1.790 for theta, with intervals $[1.710, 1.790]$ and $[1.737, 1.849]$. The sphere's median number of position queries falls from 3872 to 2304, a reduction of 40.5%. Under the prefix envelope, sphere is faster than tree, with paired sphere/tree ratio 0.916 and interval $[0.907, 0.926]$. The common reduction across evaluators follows from the tighter bound on unvisited posterior mass.

Grid refinement gives a separate measure of spatial resolution. Refining 25 to 12.5 m changes the paired position means by 2.085 m RMS, while localization RMSE changes from 14.027 to 13.925 m. Refining again to 6.25 m reduces the mean change to 0.1188 m RMS. The numerical allowance for the latter comparison is 6.09×10^{-6} m, which distinguishes summation error from the observed grid effect.

B. Calibration Information and Localization Accuracy

The localization study uses four sources and periods $c/[30000(12, 4, 24, 6)]$. Each of seven conditions contains 1000 independent trials. The position prior is the cosine-squared density of Section V, centered at $(250, 750)$ m with $w_1 = w_2 = 3500$ m. All methods use the same observations, prior, 6.25-m grid, and TV tolerance 10^{-5}. The nominal covariance is (61); the separate perturbations increase fine noise, increase calibration variance, add a Gaussian quadratic-calibration coefficient with 60-m standard deviation, change the true source correlation to 0.85 while retaining 0.35 in inference, or introduce covariance-preserving Student-t_4 measurement noise.

Table II shows the localization benefit of the finite calibration prior. In the nominal condition, marginal inference reduces RMSE from 20.92 m for source differences to 14.87 m, or 28.9%. The paired MSE reduction is 216.62 m^2, with interval $[67.06, 489.39]$. Treating calibration errors as independent while retaining their marginal variances gives 31.30 m RMSE. A constant-bias model gives 15.58 m, and diagonalizing the measurement noise gives 15.37 m. The shared frequency basis therefore accounts for the principal improvement over independent calibration errors.

The marginal and maximum-component curves nearly coincide in Fig. 5. Across all seven conditions, their paired MSE intervals, including the saved numerical allowances, contain zero. The largest estimated difference is 16.56 m^2 at 150-m fine noise, with interval

TABLE II
POSITION RMSE IN METRES FOR 1000 PAIRED TRIALS PER CONDITION. MAXIMUM USES JOINT INTEGER MAXIMIZATION AND A NORMALIZED POSITION MEAN; DIFFERENCES USE (18).

Condition	Marginal	Maximum	Differences	Coarse
Nominal	14.87	14.87	20.92	59.72
Fine noise 60 m	26.07	26.08	36.57	58.77
Fine noise 150 m	49.88	50.05	75.04	59.17
Large calibration	15.97	15.97	21.73	74.24
Quadratic response	24.75	24.75	27.67	62.92
Correlation 0.85	8.36	8.36	7.36	47.53
Student-t_4	14.78	14.78	29.52	55.35

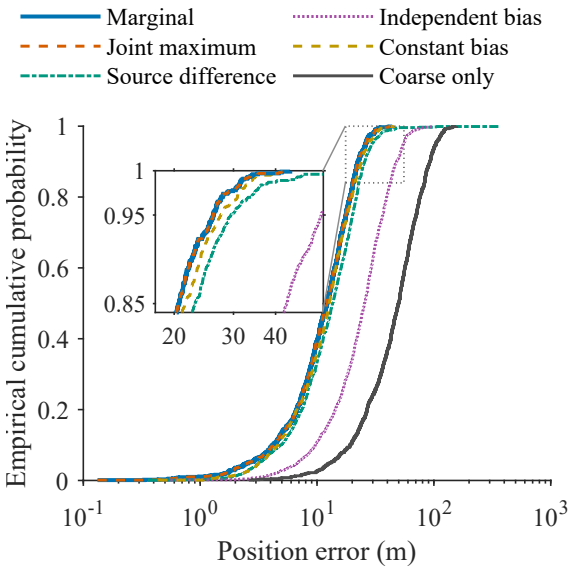
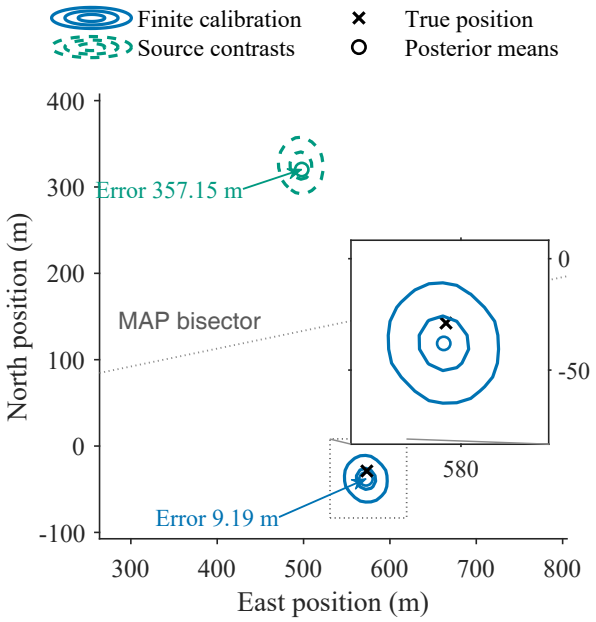


Fig. 5. Nominal position-error distributions over 1000 common trials. The inset resolves the upper part of the distributions for marginal inference, joint maximum components, and source differences. The independent-bias and coarse-only curves show the effect of the calibration model and fine measurements, respectively.

$[-12.67, 44.76]$. In these measurement regimes, one joint integer component generally dominates and the calibration model explains the observed accuracy gain.

Figure 6 illustrates the mechanism in (24). Source differences produce a 357.15-m error, whereas the finite-prior mean has a 9.19-m error. The competing alias requires calibration slopes from -188 to -273 m against a 25-m prior standard deviation. Its log preference changes from $+4.330$ under flat calibration to -112.029 with the finite prior. The prior thus suppresses an alias that fits the retained contrasts but requires a large receiver-response displacement. Under Student-t_4 noise, errors above 100 m occur in 8/1000 source-difference trials, 0/1000 marginal trials, and 55/1000 coarse-only trials; the zero-count Wilson upper limit is 0.00383.

Model sensitivity appears in both point estimates and posterior coverage. With source correlation 0.85,



Contours: 50% and 95% HPD

Fig. 6. Effect of calibration information in nominal trial 477. Contours enclose 50% and 95% posterior mass. The source-contrast solution selects a distant ambiguity, while the finite calibration prior concentrates the posterior near the true position. The inset enlarges the latter mode and its posterior mean.

source differences achieve 7.36 m RMSE and marginal inference with the assumed covariance gives 8.36 m; using the correct covariance gives 7.12 m. The paired difference-minus-marginal MSE interval, enlarged for numerical error, is $[-20.96, -10.62]$ m^2. Under the quadratic response, 95% highest-posterior-density coverage is 0.672 for marginal inference and 0.658 for differences, increasing to 0.958 with the correct response model.

The Bayesian information calculation provides a further comparison. For the correctly specified quadratic-response model, the 1000-trial sample RMSE is 21.27 m and the estimated Van Trees value is 22.31 m. A second independent 1000-trial sample gives 22.58 m RMSE, with interval $[21.83, 23.34]$; its mean squared loss minus posterior covariance trace is 8.50 m^2, with interval $[-24.94, 43.48]$. Paired refinement of the first 100 trials in each main condition to 3.125-m spacing gives a maximum mean-change RMS of 0.00205 m; all 56 method-condition resolution criteria are satisfied.

C. Finite-Waveform Localization

1) Signal generation and processing: The finite records in (2) use orthogonal frequency-division mul-

tiplexing (OFDM) with 52 resource blocks, 30-kHz subcarrier spacing, a 1024-point FFT, and a 30.72-MS/s sample rate per stream. Each 0.5-ms burst has 14 symbols and 8736 occupied resource elements. A known 127-element primary synchronization signal (PSS) supports coarse acquisition; the remaining resources contain configured pilots and unknown QPSK payload. Table III specifies the one-symbol fine measurements. Sources 1–3 use channel-state information reference signals (CSI-RS), with rows 2, 1, and 2. Source 4 uses the first of six differently offset positioning reference signal (PRS) symbols [18]; their coherent union gives a 9993.08-m period.

The source and receiver coordinates remain those of Fig. 1; positions now have a uniform prior on $[100, 400] \times [600, 900]$ m, representing an external position cue. Receiver calibration uses $f_* = 3.39$ GHz, $F_* = 90$ MHz, and coefficient standard deviations $(3, 1)$ m. The polynomial group delay

$$b_n(f) = \beta_{n0} + \beta_{n1}\xi + \gamma_n\xi^2, \qquad \xi = (f - f_*)/F_*, \tag{62}$$

is implemented through the phase response of the entire sampled waveform. Nominally $\gamma_n = 0$, reference SNR is 10 dB, and the target gain has unit magnitude. Surveillance noise variance is $P_m 10^{-\mathrm{SNR}/10}$, where P_m is the mean signal power over the active burst. The reported SNR is therefore received burst SNR per sample.

Four separate perturbations introduce a 15-m standard deviation for γ_n, a residual frequency offset $1500\zeta_n f_m/f_*$ Hz with equiprobable receiver signs $\zeta_n \in \{-1, 1\}$, direct-path amplitude 10 relative to the target, or reference SNR -15 dB. Each retains the other nominal settings. The direct-path epoch is randomized over 64–192 sample intervals and acquired over a fixed $[0, 256]$-sample search. The resulting estimate and the external position region determine the echo search, including a 50-m calibration margin.

PSS correlations yield the coarse range and the FFT-window locations. Within each selected pilot set $\mathcal{P}$, elimination of the unknown complex gain gives the delay profile

$$J(\delta) = \frac{\left|\sum_{k\in\mathcal{P}} a_k^* Y_k e^{\mathrm{i}2\pi k\Delta f\delta}\right|^2}{\sum_{k\in\mathcal{P}} |a_k|^2}. \tag{63}$$

The reference–surveillance peak difference, corrected for the window origins and reduced modulo L_m, gives y_{nm}. This profile describes a frequency-compensated, locally delay-like channel in a suitably placed cyclic-prefix window. Timing, dispersion, and ambiguity errors are retained in the evaluated trials.

Independent training uses 300 nominal trials at each SNR to fit observation means and covariance parameters after removing the known training geometry and calibration. Circular concentration determines the periodic scales, and principal residuals determine their cross-correlations. These fits are frozen before the 500 test trials at each of -20, -10, and 0 dB. The generator SNR selects the corresponding fit. All methods share each trial's target, waveform, calibration, and noise realization.

TABLE III
FINE-RANGING RESOURCES AND PERIODS FOR THE FOUR OFDM ILLUMINATORS. THE PILOT COUNT REFERS TO THE SYMBOL USED FOR THE COMPRESSED FINE OBSERVATION.

Source	Carrier (GHz)	Comb K_m	Pilots	L_m (m)
1	3.30	12	52	832.76
2	3.36	4	156	2498.27
3	3.42	24	26	416.38
4	3.48	6	104	1665.51

Two additional front ends assess the information available beyond one-symbol pilot peaks. The all-pilot method uses every configured pilot symbol. The full-record cross-ambiguity function (CAF) uses the measured noisy reference, including the unknown payload [19]. Its nominal search fixes frequency at zero; the offset-aware comparison searches a prescribed ± 3-kHz interval. CAF position estimates minimize a correlated Gaussian range objective with the same position region and calibration information. A 7×7 start grid supplies the best nine initial points for box-constrained Gauss–Newton iterations, with a projected-stationarity threshold of 10^{-5} m. The CAF estimates are modes of this range objective, while compressed-observation estimates are posterior means.

All 79500 applicable position outputs attain their numerical or solver stopping criterion. The marginal and maximum-component calculations use TV tolerance 10^{-5}. Main grid spacings are 5 m at -20 dB and 0.0625–0.5 m at the two higher SNRs. Halving the higher-SNR grids for the first 100 test trials satisfies all 70 criteria: the paired mean-change RMS, including numerical allowances, is below both 0.05 m and 5% of the finer-grid RMSE. Its largest adjusted value is 0.03536 m. These checks quantify resolution sensitivity of the specified grids.

2) Pilot statistics and position estimates: Table IV summarizes the compressed-measurement and full-record results. At nominal 0 dB, marginal fusion gives 0.44-m RMSE, compared with 0.50 m for source differences and 1.08 m for coarse ranges. At -10 dB, the ordering changes: coarse-only inference gives 6.70 m and marginal fusion gives 12.57 m. The paired coarse-minus-marginal MSE is -113.19 m^2, with numerical interval $[-274.39, -10.05]$.

Figure 7 locates this change in the upper error tail. The training residuals have a narrow center and occasional

TABLE IV
WAVEFORM LOCALIZATION RMSE IN METRES. EACH ROW USES 500 PAIRED TRIALS. THE CAF COLUMN USES THE FULL MEASURED REFERENCE, WITH ZERO-FREQUENCY SEARCH AND WITHOUT DIRECT-PATH PROJECTION.

Condition	SNR (dB)	Marginal	Differences	Coarse	CAF
Nominal	−20	123.02	123.82	124.69	0.40
	−10	12.57	18.88	6.70	0.14
	0	0.44	0.50	1.08	0.05
Quadratic	−20	124.07	123.80	125.20	5.45
	−10	13.46	22.14	14.73	5.60
	0	5.42	5.62	5.03	5.62
Offset	−20	124.03	123.73	124.96	177.86
	−10	17.36	22.67	21.55	0.46
	0	0.46	0.53	1.10	0.16
Direct path	−20	125.22	124.00	124.03	227.33
	−10	236.22	244.26	242.36	242.51
	0	243.67	243.65	192.32	242.51
Weak reference	−20	124.73	123.91	125.04	238.09
	−10	140.45	161.40	134.90	10.67
	0	160.87	169.22	165.07	0.49

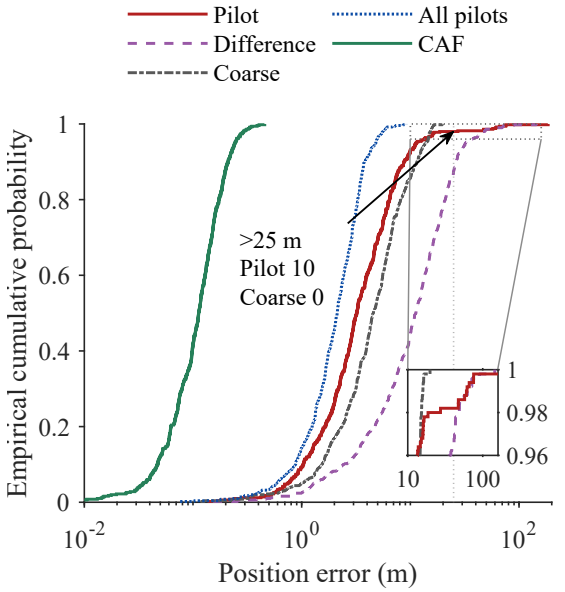


Fig. 7. Position-error distributions from 500 nominal waveform trials at −10 dB. "Pilot" is marginal inference from coarse and single-symbol fine measurements. The inset identifies the pilot tail: 10 errors exceed 25 m, compared with zero for coarse-only inference. The all-pilot and CAF curves use progressively more of the available record.

distant peaks: for source 2 at −10 dB, the median absolute principal residual is 0.862 m, while one 912.395-m return contributes 99.759% of squared residual energy. The fitted circular scale is 20.976 m and the principal-residual RMS is 26.370 m. The Gaussian compression assigns these events according to their circular concentration and covariance, whereas their occurrence is governed by peak selection in the finite record. The comparison with (56) identifies this compression as a material part of the localization problem.

At −20 dB, marginal and maximum-component RMSEs are 123.02 and 125.39 m. Their paired MSE difference is 588.54 m^2, with interval $[169.65, 983.73]$, and marginal inference retains 492/500 errors above 25 m. At −10 dB, the corresponding RMSEs are 12.57 and 12.63 m, with paired MSE difference 1.41 m^2 and numerical interval $[0.89, 1.98]$. Marginalization helps when several integer components contribute, while the quality of the extracted measurements determines the resulting localization scale.

Using all configured pilots reduces the nominal −10-dB RMSE to 2.65 m, and full-record CAF gives 0.14 m. With the weak reference at 0 dB, the compressed principal-residual RMS is 270.98 m across 8000 links, compared with 0.73 m for CAF ranges after removal of geometry and calibration in evaluation. These comparisons measure the value of retained waveform information as well as the choice of position estimator.

3) Receiver impairments and interference processing: The impairment comparisons connect improvements to the physical terms in (2). With quadratic channel response at 0 dB, incorporating its known coefficient variance reduces CAF position RMSE from 5.62 to 1.25 m. Under residual frequency offset at −20 dB, joint delay–frequency search reduces CAF RMSE from 177.86 to 0.39 m, and errors above 25 m decrease from 396/500 to 0/500. The zero-count Wilson upper limit is 0.00762.

Direct-path interference produces a different error mechanism. At 0 dB, 7922 of 8000 coarse peaks are closer to the direct path than to the echo. A nine-column cancellation subspace restores useful CAF measurements, although near-direct echoes can lose substantial energy. In one saved case, projection retains 12.31% of the target energy. We also apply the amplitude-profile normalization of [4], which accounts for the projected template energy when comparing candidate delays.

Figure 8 shows the main improvement from cancellation and the additional change from profiling. RMSE decreases from 242.51 m for the raw CAF to 12.66 m after cancellation and 12.02 m with normalized profiling. The latter paired MSE change is -15.66 m^2, with interval $[-24.60, -6.29]$, and the 95th percentile decreases from 18.37 to 17.10 m. This profile comparison reuses the 500 test records, including the case used to develop the extension. At −20 and −10 dB, profiling also lowers RMSE, from 10.46/12.25 to 9.98/11.76 m, while the 99th percentiles change from 20.60/21.66 to 20.99/22.36 m. The distributional comparison identifies both the average improvement and the remaining tail behavior.

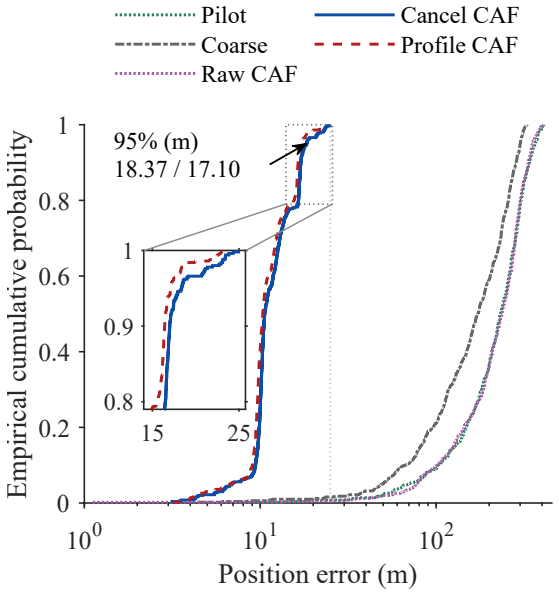


Fig. 8. Localization with direct-path interference at 0 dB. Cancellation separates the echo from the dominant direct component. The inset compares cancellation and normalized-profile CAF near the upper tail; their 95th-percentile position errors are 18.37 and 17.10 m. All curves contain the same 500 trials.

VII. Conclusion

This work has developed Bayesian position inference for passive radar measurements with a shared frequency-dependent receiver response and source-dependent ambiguity periods. Integrating the calibration coefficients produces correlated periodic receiver likelihoods. The source-contrast analysis shows which geometric directions disappear when calibration is removed, and the component decomposition explains how a finite prior resolves some of the remaining position ambiguities.

Positive bounds on integer branches and unevaluated positions provide a common stopping criterion for posterior computation. The resulting error bound applies to both the finite-grid distribution and its position mean, and it can be used with conditional-tree, sphere, or theta evaluation. The timing results show that the observation-dependent envelope provides savings across evaluators, while covariance structure determines the most efficient inner calculation.

The localization results establish a separate role for the measurement model. Calibration information improves accuracy relative to invariant source differences in the nominal setting, whereas the finite-waveform study links the achieved performance to the resources retained by the front end, the reference quality, and the receiver impairments. These findings motivate extensions that retain richer delay likelihoods and propagate calibration information across successive observations while preserving explicit posterior error control.

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